D H A R

Volume II — Engineering Roadmap

STAGE-ALPHA

“Forge of Dhar — a bridge built to be crossed once, then burned.”

Document	Stage-alpha — Forge of Dhar
Version	1.0
Date	September 6, 2026
Status	Ratified
Part of	The Canon of Dhar, Volume II
Depends on	Spec0, Spec1
Precedes	Stage-beta — Honing Dhar

1. Mandate

Stage-alpha's job is singular and disposable: prove the full compilation pipeline once, end to end, for the smallest subset of Dhar capable of expressing Stage-beta's own source code. It is not the final compiler architecture, is not required to be fast, elegant, or complete beyond that subset, and is explicitly permitted — expected — to be retired once it has done this one job.

2. Scope

2.1 In scope

- Tokenizing `.dhar` source text.
- A recursive-descent parser (Spec0 §3, pillar 1) covering exactly the bootstrap subset defined in §2.2.
- A symbol table sufficient for the bootstrap subset's scoping rules.
- Direct x86-64 assembly emission — targeting one architecture only.
- File I/O sufficient to read source and write a native output binary.

2.2 The bootstrap subset

Stage-alpha **must** support, at minimum:

Category	Required constructs
Bindings	`lock`, `flux`
Functions	`task`, `give`
Branching	`when`, `shift`, `fallback`
Loops	`span`, `cycle`, `escape`, `skip`
Data	`mold` (layout only), array indexing, `i32`, `str`, `bool`, `raw[N]`
Modules & I/O	`pull`, `expose`, `gen`

`forge`, the full 38-keyword surface, collections beyond array indexing, `wild`, and every construct in Spec1 not listed above are explicitly **not required** to work in Stage-alpha, and Stage-alpha **must** reject them with a clear diagnostic rather than mishandling them silently.

2.3 Out of scope

- The Universal IR and multi-backend architecture (Spec0 §4) — Stage-alpha targets x86-64 directly and only; the IR boundary begins at Stage-beta.
- Any optimization whatsoever.
- Self-hosting — Stage-alpha is, by definition, not written in Dhar.

3. Required Architecture

Stage-alpha's source **must** be organized as one-definition-per-file modules from its first commit, not refactored into this shape later. Every label and data symbol **must** exist in exactly one file.

Module	Responsibility
data.asm	Global data, keyword and string tables
entry.asm	Program entry point only — no other logic
tokenize.asm	Character scanning, token creation
parse.asm	Recursive-descent parsing per §2.2
symbols.asm	Symbol-table management
frames.asm	Stack-frame analysis
calls.asm	Function-call handling
codegen.asm	Code generation via the template-emitter approach (Spec0 §3, pillar 1)
runtime.asm	Minimal runtime support, error reporting

The main entry file **must** consist only of `%include` directives and the program entry point — no routine may be defined there. This rule exists because splitting a monolith into modules only counts once nothing is left duplicated between the split files and the original: a module list is not modularity if the old file still contains everything it was split out of.

4. Acceptance Criteria

1. Assembles and links into a native executable using only standard toolchain calls.
2. Correctly tokenizes, parses, and compiles every construct in the bootstrap subset (§2.2), verified by at least one passing test per construct.
3. Rejects, with a clear diagnostic, any construct outside the bootstrap subset.
4. Successfully compiles Stage-beta's complete source tree end to end, producing a working Stage-beta executable.
5. Produces reproducible builds: two clean builds from the same source are byte-identical, ignoring timestamps.

5. Handoff to Stage-beta

```
Stage-alpha compiles Stage-beta's source
        |
        v
Stage-beta executable (generation 1)
```

Once this handoff succeeds and Stage-beta separately passes its own self-hosting acceptance criteria, Stage-alpha's mandate is complete. It **may** then be archived as a historical bootstrap artifact per Spec0 §3, pillar 5. It **must not** be extended with new features after this point — any new capability belongs in Stage-beta or later, never retrofitted into the bridge.

Stage-alpha — Forge of Dhar · v1.0 · September 6, 2026 · Ratified. Revision log: v1.0 — first ratified edition.